Activity 2: Simple House Price Prediction Using Linear Regression

1. Aim

To develop a Linear Regression model that predicts house prices based on the area of the house.

2. Dataset Description

- Dataset: Housing Price Prediction
- Number of records: 545 (before cleaning)
- Number of columns: 13
- Input feature: Area
- Target variable: Price

3. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the housing dataset using Pandas.
2. Checked missing values and duplicate records.
3. Removed duplicate records.
4. Removed rows with missing area or price.
5. Removed invalid zero or negative values.

4. Data Visualization

A scatter plot was created to understand the relationship between house area and house price.

5. Model Training

- Algorithm: Linear Regression
- Training data: 80%
- Testing data: 20%
- Random state: 42

6. Linear Regression Equation

The equation is:

Price = b0 + (b1 × Area)

- Intercept (b0): [Enter your value]
- Coefficient (b1): [Enter your value]

7. Model Evaluation

Metric	Result
MAE	[Enter value]
MSE	[Enter value]
RMSE	[Enter value]
R^2 Score	[Enter value]

8. Actual vs Predicted Visualization

An Actual vs Predicted scatter plot was created to compare the actual house prices with the prices predicted by the model.

9. Prediction for a New House

- New house area: [Enter your value]
- Predicted house price: [Enter your value]

10. Conclusion

A Linear Regression model was successfully developed to predict house prices based on house area. The dataset was inspected and cleaned before training. The model was evaluated using MAE, MSE, RMSE and R^2 score. The actual and predicted prices were visualized to understand model performance.